

1.6 - Expansion of the Wakes

In practice we are mostly interested in knowing the wake functions in a region very close to the vacuum chamber's center - where the beam travels. In that region $\vec{\rho}_s$ and $\vec{\rho}_w$ are much smaller than the transverse dimensions of the chamber, allowing us to expand W in its transverse coordinates and keep only low-order terms:

$$\begin{aligned} W(\vec{\rho}_s, \vec{\rho}_w, z) &\approx W_0 + \vec{\rho}_s^T \cdot \vec{M}_s + \vec{\rho}_w^T \cdot \left(\vec{M}_w + D \cdot \vec{\rho}_s + \frac{1}{2} Q \cdot \vec{\rho}_w \right) = \\ &= W_0 + \underbrace{\vec{\rho}_s^T \cdot \vec{M}_s + \vec{\rho}_w^T \cdot \vec{M}_w}_{\text{monopole terms}} + \underbrace{\vec{\rho}_w^T \cdot D \cdot \vec{\rho}_s}_{\text{Dipole term}} + \underbrace{\frac{1}{2} \vec{\rho}_w^T \cdot Q \cdot \vec{\rho}_w}_{\text{Quadrupole term}} + \frac{1}{2} \vec{\rho}_s^T \cdot Q \cdot \vec{\rho}_s + \dots \end{aligned}$$

↳ less relevant

where W_0 , M_s , M_w , D and Q are dependent on z

Recovering the wake functions (until 1st order on $\vec{\rho}_s$ and $\vec{\rho}_w$)

$$\bullet \quad w_{||} = \frac{\partial W}{\partial z} \approx \underbrace{W_0}_{\text{usually the only term considered for } w_{||}} + \vec{\rho}_s^T \cdot \vec{M}_s + \vec{\rho}_w^T \cdot \vec{M}_w + \dots$$

* ↳ usually the only term considered for $w_{||}$

dominates the expansion ~ usually referred to as the longitudinal wake
→ responsible for longitudinal effects:

- ▼ avg. energy loss
- ▼ potential-well distortion
- ▼ bunch-lengthening/shortening
- ▼ coupled bunch oscillations

$$\bullet \quad \vec{w}_\perp = \vec{\nabla}_{w\perp} W \approx \vec{M}_w + D \cdot \vec{\rho}_s + Q \cdot \vec{\rho}_w$$

* $\vec{M}_s$ and $\vec{M}_w$ ~ usually referred to as monopolar wakes → effect independent of the transv. displacement of the source and with particles
↳ change in the closed orbit ~ does not lead to instabilities
as a func. of the current

→ in most practical cases these terms are zero due to symmetries in the vacuum chamber

★ $\mathcal{D} = \begin{pmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{pmatrix} \sim D_{ij}'$ are referred to as dipolar wakes ~ as they are originated from an offset of the source particle from the chamber's center, and

↳ $D_{11} \sim$ horizontal dipolar wake

↳ $D_{22} \sim$ vertical dipolar wake

Generates a wake:

$$\vec{w}_1^D = \mathcal{D} \cdot \vec{\rho}_s, \text{ which is thus independent from } \vec{\rho}_w,$$

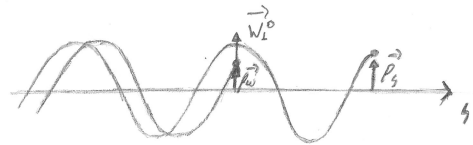
Just like in the case of a mag. dipole

generate a wake that is "dipole-like", in the sense that it doesn't depend on the coord.'s x and y of the witness particle!

↳ Responsible for the transverse oscillations that cause coupled bunch oscillations, emittance deterioration and even beam loss

* These terms are particularly dangerous as they describe a mechanism where oscillations of the source particle can induce oscillations on the witness particle, which always leads to a resonant effect

as both particles naturally already oscillate with very similar frequencies (equiv. to saying they have approx. the same tune)



→ $D_{12} = D_{21}$ is usually zero due to symmetries in the vacuum chamber, and even when it is not, its effects are not significant as the hor. and vert. tunes are very different, and thus no resonant effects appear - oscillations in one plane do not transfer to oscillations in the other.

★ Q_{11} and $Q_{22} \sim$ Detuning or Quadrupolar wakes

$\vec{w}_1^Q = Q \cdot \vec{\rho}_w \rightarrow$ Wake/kick whose strength depends on the transv. coordinates of the witness particle itself → just like in a quadrupole magnet

Furthermore, if $v \approx c$, $Q_{22} = -Q_{11}$, producing opposite focalizations on x and y , precisely as the mag. quad. does → generates a focusing effect

$\sim Q_{12} = Q_{21}$ usually = 0 from symmetry considerations

Further understanding the symmetry implications and simplifications that become evident from the wake potential $W(\vec{\rho}_s, \vec{\rho}_w, z)$ formalism (highlighting its power):

Again,

$$W(\vec{\rho}_s, \vec{\rho}_w, z) = W_0 + \vec{\rho}_s^T \cdot \vec{M}_s + \vec{\rho}_w^T \cdot \vec{M}_w + \vec{\rho}_w^T \cdot \mathcal{D} \cdot \vec{\rho}_s + \frac{1}{2} \vec{\rho}_w^T \cdot \mathcal{Q} \cdot \vec{\rho}_w$$

Now, let's consider a chamber with mirror symmetry (e.g. rectangular or elliptical)
($x \rightarrow -x$ and/or $y \rightarrow -y$ do not change W)

Take for instance $x \rightarrow -x$: (exception example: radiation masks)

$$W(-x_s, y_s, -x_w, y_w, z) = W(+x_s, y_s, +x_w, y_w, z)$$

$$\bullet (-x_s \ y_s) \cdot \begin{pmatrix} M_{sx} \\ M_{sy} \end{pmatrix} = (x_s \ y_s) \cdot \begin{pmatrix} M_{sx} \\ M_{sy} \end{pmatrix} \Rightarrow -M_{sx} = M_{sx} \Rightarrow M_{sx} = 0$$

• Analogously for $\vec{\rho}_w^T \cdot \vec{M}_w$, we get $M_{wx} = 0$

† Doing the same but with $y \rightarrow -y$, we get: $M_{sy} = 0$, $M_{wy} = 0$ *

$$\bullet (-x_w \ y_w) \cdot \begin{pmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{pmatrix} \cdot \begin{pmatrix} -x_s \\ y_s \end{pmatrix} = (-x_w \ y_w) \cdot \begin{pmatrix} -D_{11}x_s + D_{12}y_s \\ -D_{21}x_s + D_{22}y_s \end{pmatrix} = D_{11}x_w x_s - D_{12}x_w y_s - D_{21}x_s y_w + D_{22}y_s y_w,$$

$$\text{while } (x_w \ y_w) \cdot \mathcal{D} \cdot \begin{pmatrix} x_s \\ y_s \end{pmatrix} = D_{11}x_s x_w + D_{12}x_s y_w + D_{21}x_w y_s + D_{22}y_s y_w,$$

and thus, their equality for any x_s, y_s, x_w, y_w demands:

$$\begin{cases} -D_{12} = D_{12} \Rightarrow D_{12} = 0 \\ -D_{21} = D_{21} \Rightarrow D_{21} = 0 \end{cases}$$

• Proceeding analogously for $\vec{\rho}_w^T \cdot \mathcal{Q} \cdot \vec{\rho}_w$, we find $Q_{12} = Q_{21} = 0$

$\Rightarrow$ Non-diagonal terms of $\mathcal{D}$ and $\mathcal{Q}$ are all zero!

Furthermore, since W is an harmonic function on x_w and y_w for ultra-relativistic electrons,

$$\nabla_{w\perp}^2 W = \frac{\partial^2 W}{\partial x_w^2} + \frac{\partial^2 W}{\partial y_w^2} = 0, \text{ and the quadratic terms with } x_w \text{ and } y_w \text{ on the } W$$

expansion (which can thus "survive" two "witness-transverse" derivatives) are:

$$2 \cdot W^{\text{quad}} = \vec{\rho}_w^T \cdot Q \cdot \vec{\rho}_w = Q_{11} x_w^2 + 2 Q_{12} x_w y_w + Q_{22} y_w^2, \text{ we have:}$$

$$\frac{\partial^2 W}{\partial x_w^2} + \frac{\partial^2 W}{\partial y_w^2} = Q_{11} + Q_{22} = 0 \Rightarrow \boxed{Q_{11} = -Q_{22}}$$

And thus for a vacuum chamber w/ mirror symmetry containing ultra-rel electrons:

$$\begin{aligned} \rightarrow \vec{M}_b = \vec{M}_w = \vec{0} \quad & \rightarrow D = \begin{pmatrix} D_{11} & 0 \\ 0 & D_{22} \end{pmatrix}; \quad \rightarrow Q = \begin{pmatrix} -Q_{22} & 0 \\ 0 & Q_{22} \end{pmatrix} \end{aligned}$$

and W 's expansion simplifies to:

$$W(\vec{\rho}_s, \vec{\rho}_w, z) = W_0 + D_{11} x_s x_s + D_{22} y_w y_s - \frac{1}{2} Q_{22} x_w^2 + \frac{1}{2} Q_{22} y_w^2 //$$

W_0, D_{11}, D_{22} and Q_{22} all functions of z .

This enables recovering the recurrent wake functions found in the literature:

defining $W_x^D \triangleq D_{11}$, $W_y^D = D_{22}$, and $W^Q = Q_{22}$, we get:

$$\left\{ \begin{aligned} \bullet W_{11}(z) &= \frac{\partial W}{\partial z} = W_0'(z) \\ \bullet W_x(x_s, x_w, z) &= W_x^D(z) \cdot x_s - W^Q(z) \cdot x_w \\ \bullet W_y(y_s, y_w, z) &= W_y^D(z) \cdot y_s + W^Q(z) \cdot y_w // \end{aligned} \right.$$

To determine such terms, one can first do a simulation with $x_s \neq 0, x_w = 0$ to find $W_x^D (= w_x/x_s)$, and then perform another sim. w/ $x_s = 0, x_w \neq 0$ to get $W^Q (= -w_x/x_w)$!

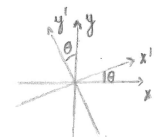
Lastly, if the vacuum chamber presents not only mirror symmetry, but also

rotation symmetry — i.e., cylindrical symmetry — W must also satisfy:

(exception example: undulators)

$$W(R \cdot \vec{\rho}_s, R \cdot \vec{\rho}_w, z) = W(\vec{\rho}_s, \vec{\rho}_w, z),$$

where R is a rotation matrix $\left(\begin{smallmatrix} \oplus \\ R = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \end{smallmatrix} \right)$



$$\begin{aligned} x' &= x \cos \theta + y \sin \theta \\ y' &= -x \sin \theta + y \cos \theta \\ \Rightarrow \begin{pmatrix} x' \\ y' \end{pmatrix} &= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \end{aligned}$$

Plugging into the wake expansion:

$$W(R \cdot \vec{\rho}_s, R \cdot \vec{\rho}_w, z) = W_0 + \vec{\rho}_w^T (R^T \cdot D \cdot R) \vec{\rho}_s + \vec{\rho}_w^T (R^T \cdot Q \cdot R) \vec{\rho}_w$$

And thus we need:

$$\begin{aligned} \bullet R^T \cdot D \cdot R &= D \Rightarrow \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} D_{11} & 0 \\ 0 & D_{22} \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} D_{11} & 0 \\ 0 & D_{22} \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} D_{11} \cos \theta & D_{11} \sin \theta \\ -D_{22} \sin \theta & D_{22} \cos \theta \end{pmatrix} = \begin{pmatrix} D_{11} \cos^2 \theta + D_{22} \sin^2 \theta & (D_{11} - D_{22}) \sin \theta \cos \theta \\ (D_{11} - D_{22}) \sin \theta \cos \theta & D_{11} \sin^2 \theta + D_{22} \cos^2 \theta \end{pmatrix} = \end{aligned}$$

which can only be true if: $D_{11} = D_{22}$, i.e., $W_x^D = W_y^D = W^D$

• Analogously, $R^T \cdot Q \cdot R = Q \Rightarrow Q_{11} = Q_{22}$, but from $\nabla_{\perp}^2 W = 0$ we have $Q_{11} = -Q_{22}$,

leaving as the only option: $Q_{11} = Q_{22} = 0 \Rightarrow W^Q = 0 \Rightarrow$ No quadrupolar wake!

$\Rightarrow$ Therefore, for a perfectly round chamber, $\vec{M}_s = \vec{M}_w = 0$; $D = \begin{pmatrix} W^D & 0 \\ 0 & W^D \end{pmatrix} = W^D \cdot \mathbb{I}$; $Q = 0$, and

the wakes become:

$$\begin{aligned} \bullet W_H(z) &= W_0(z) \\ \bullet W_x(x_s, z) &= W^D \cdot x_s \\ \bullet W_y(y_s, z) &= W^D \cdot y_s \end{aligned} \left\{ \vec{W}_{\perp} = W^D \cdot \vec{\rho}_s \right.$$

Thus, in short:

♦ General case:

$$\bullet w_{||}(z) = W_0'(z)$$

$$\bullet \vec{w}_{\perp}(\vec{\rho}_s, \vec{\rho}_w, z) = \vec{M}_w(z) + D \cdot \vec{\rho}_s + Q \cdot \vec{\rho}_w$$

♦ Mirror-symmetry

$$\bullet w_{||}(z) = W_0'(z)$$

$$\bullet w_x(x_s, x_w, z) = W_x^D(z) \cdot x_s - W_x^Q(z) x_w$$

$$\bullet w_y(y_s, y_w, z) = W_y^D(z) y_s + W_y^Q(z) y_w$$

♦ Axisymmetry

$$\bullet w_{||}(z) = W_0'(z)$$

$$\bullet w_x(x_s, z) = W^D(z) \cdot x_s$$

$$\bullet w_y(y_s, z) = W^D(z) \cdot y_s$$

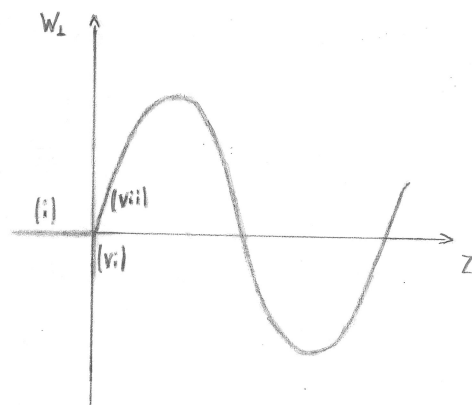
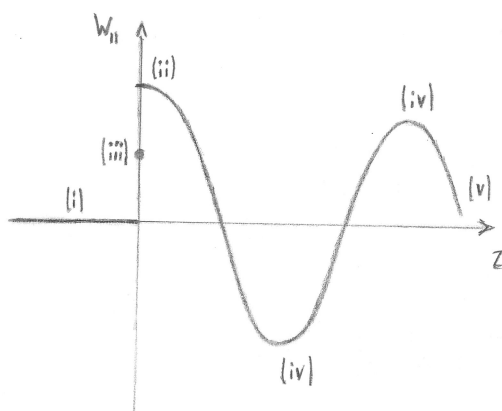
1.7 - Properties of the wakes

* The next sections are based on the books on collective effects from A. W. Chao; K. Y. Ng; and S. Khan.

As discussed previously, for ultra-relativistic particles the wakes satisfy the "Causality Principle", as they travel at the same speed of the particles, c , and thus cannot "catch up" with a particle for which $z \leq 0$. In terms of the wake formalism, this means:

$$W(z \leq 0) = 0 \Rightarrow \begin{cases} \bullet \vec{w}_{\perp}(z) = \vec{\nabla}_{w\perp} W = 0, & \text{for } z \leq 0 \\ \bullet w_{||}(z) = \frac{\partial}{\partial z} W(z) = 0, & \text{for } z \leq 0 \end{cases} \quad \sim \text{considering point particles} \quad (i)$$

Furthermore, the longitudinal and transverse wakes also satisfy some additional properties, which can be essential to verify if results from simulations are physically plausible/valid. Let's draw realistic $w_{||}$, $w_{\perp}$ profiles (cylindrical symmetry for simplification, $\vec{w}_{\perp} = w_{\perp} \hat{\rho}$), and then advance to the properties that explain them!



Looking at those profiles, we see some features/properties that will always be respected by $w_{||}$ and $w_{\perp}$:

(ii) $w_{||}(0^+) \geq 0 \sim$ if $w_{||}(0^+) < 0$, a witness particle immediately behind the source particle would continuously experience an energy gain (since $w_{||} > 0 \Rightarrow$ energy loss $\sim w_{||} \propto -\Delta p_z$), which would be a direct violation of the conservation of energy

(iii) $w_{||}(0) = \frac{1}{2} w_{||}(0^+) \sim$ known as the "Fundamental Theorem of Beam Loading", it states that a charge experiences precisely half of the wake it produces on charges immediately behind it

Proof: First some background knowledge

*Adapted from [S. Khan]

From the superposition principle, the longitudinal wake "voltage" for a longitudinally extended charge distribution

$\lambda(z^*)$ is:

$$V_{||}(z) = \int_{-\infty}^{\infty} dz^* w_{||}(z^*) \lambda(z-z^*) \stackrel{\Delta}{=} q \cdot K_{||}$$

$\nearrow V$ $\nearrow m$ $\nearrow V/c$ $\nearrow c/m$ $\nearrow c$ $\nearrow V/c$
 $\nearrow$ auto-induced wake voltage

where $K_{||}$ is the longitudinal loss factor - and we're considering the charge dist. passing through cavity. With this in mind, the energy loss due to wake fields of a charge q passing cavity will be:

$$\Delta E = q \cdot V_{||}^0 = q^2 \cdot K_{||}$$

Now, for a second charge q passing just behind the 1st one, it will experience its own induced wake voltage as well as the ind. volt. left behind by the 1st (i.e., $V_{||}(0) + V_{||}(0^+)$), which we'll call $V_{||}^{0+}$. Thus, the total energy loss will be:

$$\Delta E = \Delta E_1 + \Delta E_2 = q^2 K_{||} + (q^2 K_{||} + q V_{||}^{0+})$$

However, viewing the two charges as 1 bunch of charge $2q$ implies $\Delta E = (2q) \cdot (2q \cdot K_{||}) = 4q^2 K_{||}$,

and thus:

$$2q^2 K_{||} + q V_{||}^{0+} = 4q^2 K_{||} \Rightarrow V_{||}^{0+} = 2q K_{||} = 2 \cdot V_{||}^0 \sim V_{||}^0 = \frac{1}{2} V_{||}^{0+}$$

For the next properties, it will be useful to use a multipole expansion* in cylindrical coord.'s of the longitudinal and transverse wakes (particularly useful in the case of a cylindrically symmetric chamber):

$$w_{||}(r, \theta, z) = \frac{1}{q} \sum_{m=0}^{\infty} I_m \cdot W'_m(z) \cdot r^m \cos(m\theta) ; \quad \sim -\frac{V}{q^2} \Delta p_s$$

$$w_{\perp}(r, \theta, z) = -\frac{1}{q} \sum_{m=0}^{\infty} I_m \cdot W_m(z) \cdot m \cdot r^{m-1} [\hat{r} \cos(m\theta) - \hat{\theta} \sin(m\theta)] , \quad \sim \frac{V}{q^2} \Delta p_{\perp}$$

⑧ One important remark is: $W'_m(z) \triangleq -\frac{\partial}{\partial z} W_m(z)$, due to the convention choice!

and in the case of a "cos(mθ) beam" - i.e., a beam which carries a pure m-polar moment - we have simply:

$$w_{||,m}(r, \theta, z) = \frac{1}{q} I_m \cdot W'_m(z) \cdot r^m \cos(m\theta) ; \quad w_{\perp,m}(r, \theta, z) = -\frac{1}{q} I_m \cdot W_m(z) \cdot m \cdot r^{m-1} [\hat{r} \cos(m\theta) - \hat{\theta} \sin(m\theta)]$$

↳ From these, one can understand why $W'_m(z)$ is called the longitudinal wake function of azimuthal m, while $W_m(z)$ is called the transverse wake func. of azim. m.

↳ their relation stems from the Panofsky-Wenzel Theorem, $\vec{\nabla}_{\perp} \cdot \vec{w}_{||} = \frac{\partial}{\partial z} \vec{w}_{\perp}$

→ * $W_m(z)$ describes the shock response of the vacuum chamber to a δ-function which carries on m-th moment [A.W. Chao]

* This helps us see that for a monopole beam (m=0), $w_{\perp}(z) = 0$ and $w_{||}(z) = W'_0(z) \sim \text{ind. from } \vec{r}(r, \theta)$

We are now ready! Let's proceed with the properties:

↳ precisely what we had from the transv. Taylor exp. of $W(\vec{r}_1, \vec{r}_2, z)$

$$(iv) |w_{||}(z)| \leq w_{||}(0^+), \quad z > 0$$

Proof: Let's consider two particles separated by z, with the same abs. value of multipole moment $|I_m|$.

The energy lost by a particle of mth moment I_m experiencing a wake $w_{||}(z)$ can be written: $U = I_m^2 w_{||}(z)$.

Thus, the total energy lost by the two particles is, $U_{\text{total}} = U_{\text{self},1} + U_{\text{self},2} + U_{\text{inter}} = \frac{1}{2} I_m^2 w_{||}(0^+) + \frac{1}{2} I_m^2 w_{||}(0^+) + U_{\text{inter}}$

where we used that $w_{||}(0) = \frac{1}{2} w_{||}(0^+) \sim \text{fund. theor. of beam load}$.

Furthermore, $U_{\text{inter}} = I_{m,1} \cdot I_{m,2} \cdot w_{||}(z) = \pm I_m^2 w_{||}(z)$, depending on the signs of $I_{m,i}$

Now, for a passive structure like the vacuum chamber, the beam cannot extract energy from it, and thus the total lost energy must satisfy: $U_{\text{total}} \geq 0$ (this must hold for any chosen $I_{m,1}$ and $I_{m,2}$)

Thus, if the particles have the same moment ($I_{m,2} = I_{m,1}$):

$$U_{\text{total}} = 2 \cdot \left(\frac{1}{2} I_m^2 w_{||}(0^+) \right) + I_m^2 w_{||}(z) = I_m^2 (w_{||}(0^+) + w_{||}(z)) \geq 0 \Rightarrow w_{||}(z) \geq -w_{||}(0^+)$$

And if $I_{m,2} = -I_{m,1}$:

$$U_{\text{total}} = I_m^2 w_{||}(0^+) - I_m^2 w_{||}(z) \geq 0 \Rightarrow w_{||}(z) \leq w_{||}(0^+)$$

Combining the results, we get:

$$|w_{||}(z)| \leq w_{||}(0^+)$$

■

Or, more gener., using two particles with different multipole amplitudes a and b , we have:

$$U_{\text{lost}}(a, b, z) = \frac{1}{2} a^2 w_{||}(0^+) + ab w_{||}(z) + \frac{1}{2} b^2 w_{||}(0^+) = \frac{1}{2} \underbrace{\begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} w_{||}(0^+) & w_{||}(z) \\ w_{||}(z) & w_{||}(0^+) \end{pmatrix}}_M \begin{pmatrix} a \\ b \end{pmatrix} \geq 0$$

$\Rightarrow M$ must be positive semidefinite

$$\Rightarrow w_{||}(0^+) \geq 0; \text{ and } \det(M) \geq 0 \Rightarrow w_{||}(0^+)^2 - w_{||}(z)^2 \geq 0$$

$$\Rightarrow |w_{||}(z)| \leq w_{||}(0^+)$$

■

(v) $\int_0^\infty w_{||}(z) dz \geq 0$ ~ similar proof to the previous one, but using a continuous charge distribution. If it wasn't true, one could think of a long enough beam that would extract energy from the vacuum chamber
 $\hookrightarrow$ passive structure

(vi) $w_{\perp}(0) = 0$ ~ a charge is not transversely displaced by its own wake field

(vii) $w_{\perp}(0^+) > 0$

Proof: From the multipole formalism, we've seen that $w_{||,m}(r, \theta, z) = C(r, \theta) \cdot W_m'(z)$, while

$$\vec{w}_{\perp, m}(r, \theta, z) = -(\vec{\nabla}_{\perp} C(r, \theta)) \cdot W_m(z) \stackrel{\#}{=} + \vec{\nabla}_{\perp} C(r, \theta) \int_{-\infty}^z W_m'(z^*) dz^* = + \left(\frac{I_m}{q} m r^{m-1} \right) [\hat{r} \cos m\theta - \hat{\theta} \sin m\theta] \int_0^z \left(\frac{I_m}{q} r^m \cos m\theta \right)^{-1} w_{||, m}(\dots, z^*) dz^*$$

And thus, $\vec{w}_{\perp, m} = w_{\perp, m} \cdot [\hat{r} \cos m\theta - \hat{\theta} \sin m\theta]$ with: $w_{\perp, m} = \frac{m}{r \cos m\theta} \int_0^z w_{||, m} \cdot dz^* = \begin{cases} = 0, & z = 0 \\ > 0, & z = 0^+ \text{ (from prop. iii)} \end{cases}$

Developping the $w_{||}$ properties (ii) to (vi) more uniformly and intuitively:

Remembering that $w_{||}(z) = \frac{-c}{q_1 q_2} \Delta p_s = \frac{U_{lost}}{q_1 q_2}$, then for a set of N charges q moving along

the s axis (implying a pure $m=0$ moment), the energy lost due to wake fields when passing through a cavity is:

$$U_{lost}^{total} = \underbrace{\sum_{i=1}^N q_i^2 \cdot w_{||}(0)}_{\text{"self-wake" terms}} + \underbrace{\sum_{i=1}^N \sum_{j < i} q_i q_j \cdot w_{||}(|z_j - z_i|)}_{\text{interaction terms betw. two charges}}$$

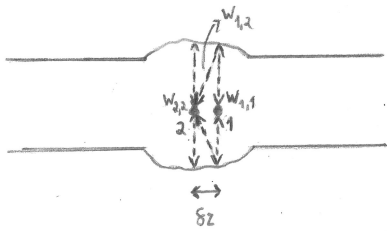
in the rigid-beam approximation, the particle's trajectories are parallel to $\hat{s}$, and thus only the longitudinal wake performs work (approximately, since in reality there are the small betatron oscillations)

In the case of off-axis charges, this generalizes by making $w_{||}(z) \rightarrow w_{||}(\vec{\rho}_i, \vec{\rho}_j, z)$ (i.e., by considering more terms in the $W(\vec{\rho}_s, \vec{\rho}_w, z)$ expansion), such that $U_{lost}^{i \rightarrow j} = q_i q_j w_{||}(\vec{\rho}_i, \vec{\rho}_j, z)$, or alternatively, by making a multipole decomposition $\sim U_{lost}^{i \rightarrow j} = \sum_m I_m^i I_m^j \cdot w_{||, m}(z)$. Nonetheless, since $\vec{\rho}_s, \vec{\rho}_w$ are typically very small, using the former generalization, $w_{||}(\vec{\rho}_s, \vec{\rho}_w, z) \approx w_{||}(z) (= W_0'(z))$, and thus the "on-axis" expression for U_{lost} suffices!

First, taking the case $N=1$ (single charge) and using $U_{lost}^{total} > 0$ (passive cavity):

$\hookrightarrow U_{lost, 1}^{total} = q^2 \cdot w_{||}(0) > 0 \Rightarrow w_{||}(0) > 0 \quad (I)$

Now, consider $N = 2$, with a small separation δz between the particles:



$$U_{\text{lost},2}^{\text{total}} = U_{\text{lost}}^{\text{self},1} + U_{\text{lost}}^{\text{self},2} + U_{\text{lost}}^{1 \rightarrow 2} =$$

$$= q^2 w_{||}(0) + q^2 w_{||}(0) + q^2 w_{||}(\delta z) \stackrel{\delta z \rightarrow 0}{=} 2q^2 w_{||}(0) + q^2 w_{||}(0^+)$$

In this $\delta z \rightarrow 0$ limit, we can see the two unit charges as a single of charge $2q$, which gives: $U_{\text{lost}}^{\text{total}} = (2q)^2 w_{||}(0) = 4q^2 w_{||}(0)$

Since the two perspectives must give the same energy loss value (for $\delta z \rightarrow 0$):

$$2q^2 w_{||}(0) + q^2 w_{||}(0^+) = 4q^2 w_{||}(0) \Rightarrow w_{||}(0^+) = 2 \cdot w_{||}(0) \geq 0 \quad (\text{I})$$

And thus:

$$w_{||}(0^+) > 0 \quad (ii); \quad w_{||}(0) = \frac{1}{2} w_{||}(0^+) \quad (iii)$$

↳ Fund. Theorem of Beam Loading

Still considering two charges, now separated by z :

$$U_{\text{lost},2}^{\text{total}} = 2q^2 w_{||}(0) + q^2 w_{||}(z) = q^2 w_{||}(0^+) + q^2 w_{||}(z)$$

$$\text{Then, } U_{\text{lost}}^{\text{total}} \geq 0 \Rightarrow q^2 (w_{||}(0^+) + w_{||}(z)) \geq 0 \Rightarrow w_{||}(z) \geq -w_{||}(0^+)$$

And even more generally, if we consider two opposite charges:

$$U_{\text{lost}}^{\text{total}} \geq 0 \Rightarrow q^2 (w_{||}(0^+) - w_{||}(z)) \geq 0 \Rightarrow w_{||}(z) \leq w_{||}(0^+)$$

Thus culminating on:

$$|w_{||}(z)| \leq w_{||}(0^+) \quad (iv)$$

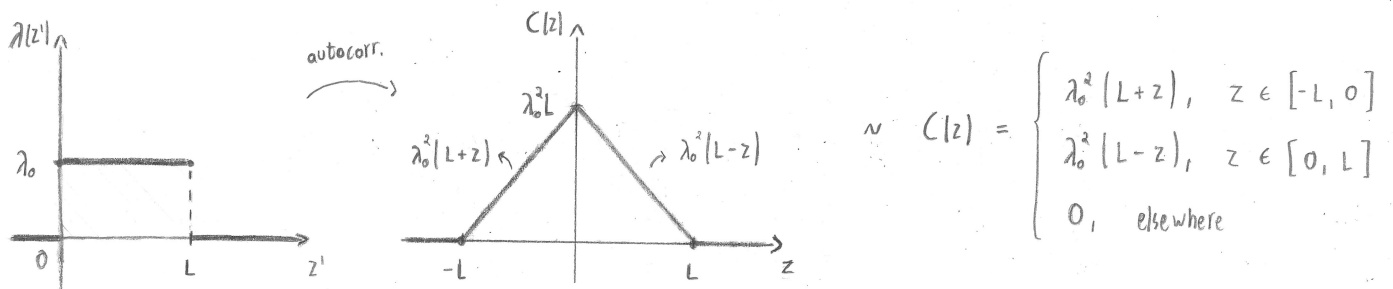
Finally, for a bunch with continuous charge distribution $\lambda(z')$, the sum of the energy losses dU_{loss} from interactions between each small source element, $dq_s = \lambda(z') dz'$, and witness element at a distance z behind it, $dq_w = \lambda(z'+z) d(z'+z)$, yields:

$$U_{\text{loss}}^{\text{total}} = \int_{-\infty}^{\infty} \int_0^{\infty} dq_s(z') dq_w(z'+z) w_{\parallel}(z) dz dz' = \int_{-\infty}^{\infty} \int_0^{\infty} \lambda(z') \lambda(z'+z) w_{\parallel}(z) dz dz'$$

Or, defining the autocorrelation:

$$C(z) \triangleq \int_{-\infty}^{\infty} \lambda(z') \lambda(z'+z) dz' \Rightarrow U_{\text{loss}}^{\text{total}} = \int_0^{\infty} C(z) \cdot w_{\parallel}(z) dz$$

Now, for a uniform charge distribution of length L , $\lambda(z') = \begin{cases} \lambda_0, & 0 \leq z' \leq L \\ 0, & \text{elsewhere} \end{cases}$



And thus:

$$U_{\text{loss}}^{\text{total}} = \int_0^{\infty} C(z) \cdot w_{\parallel}(z) dz = \lambda_0^2 \int_0^L (L-z) w_{\parallel}(z) dz \xrightarrow{\text{passivity}} \div \lambda_0^2 L \Rightarrow \int_0^L (1-z/L) \cdot w_{\parallel}(z) dz \geq 0$$

Then, taking the limit:

$$\lim_{L \rightarrow \infty} \left(\int_0^L (1-z/L) \cdot w_{\parallel}(z) dz \right) = \int_0^{\infty} w_{\parallel}(z) \cdot dz \geq 0 \quad (v)$$